Module 3

Power System Protection

Why Do We Need the Per-Unit System?

Working directly in ohms, amps and volts across a network of transformers is unwieldy

The Problem

- A power system has generators, transformers, and lines operating at many different voltage levels.
- Impedances in ohms cannot be compared or combined directly unless first referred through every transformer ratio in between.
- Manufacturers rate machines in %R and %X on their own nameplate rating — not a common base.

The Fix — Per-Unit Value

Express every quantity as a fraction of a chosen base value:

$$\text{Per unit value} = \frac{\text{actual value}}{\text{base value (same unit)}}$$

Once every quantity is on the same base, apparatus of any size can be compared, combined, and connected across transformers as one simplified network.

Base Quantities

Four base quantities are linked by Ohm's law — fixing any two fixes the rest

S_B

Base power

kVA or MVA

V_B

Base voltage

kV

I_B

Base current

A

Z_B

Base impedance

Ω (ohm)

Governing Relations (single-phase / per-phase basis)

Only two base quantities need to be chosen independently — usually base kVA and base kV. The other two then follow automatically.

$$S_B = V_B \times I_B$$

$$Z_B = V_B / I_B = (V_B)^2 / S_B$$

Base kVA and base kV are fixed independently. Base Current and base impedance are determined as follows

$$S_B = V_B \times I_B$$

So, $I_B = S_B/V_B$

Base Current

$$I_B = \frac{kVA_B}{kV_B} \quad amperes$$

$$Z_B = V_B / I_B$$

(but $I_B = S_B/V_B$)

$$Z_B = (V_B)^2 / S_B$$

Base Impedance

$$Z_B = \frac{(kV_B)^2 \times 1000}{kVA_B} \quad ohms$$

Base Values — Single-Phase Circuits

Base kVA and base kV (phase value) are chosen; current & impedance follow

Base Current

$$I_B = \frac{kVA_B}{kV_B} \quad \textit{amperes}$$

Base Impedance

$$Z_B = \frac{(kV_B)^2 \times 1000}{kVA_B} \quad \textit{ohms}$$

For single-phase, base kV is the phase (line-to-neutral) voltage and base kVA is the per-phase kVA.

Per-Unit Impedance

$$Z_{pu} = \frac{\text{Actual impedance } (\Omega)}{\text{Base impedance } Z_B}$$

Equivalently:

$$Z_{pu} = \frac{\text{Actual } Z \times kVA_B}{(kV_B)^2 \times 1000}$$

Notice %X and %R are simply p.u. × 100, evaluated at the machine's own rated base.

Base Values — Three-Phase Circuits

Line-to-line kV and total three-phase kVA are used directly as base

Base Current (line current)

$$I_B = \frac{kVA_B}{\sqrt{3} \times kV_B} \quad \textit{amperes}$$

Base Impedance (per phase)

$$Z_B = \frac{(kV_B)^2 \times 1000}{kVA_B} \quad \textit{ohms}$$

Here base kV is the line-to-line voltage and base kVA is the total 3-phase kVA — yet the impedance formula comes out identical in form to the single-phase case.

Where the √3 Goes

Starting from the star-connected phase relations,

$$Z_B = \frac{kV_B / \sqrt{3} \times 1000}{I_B}$$

substituting the line current expression and simplifying cancels the √3, leaving the same compact form as single-phase base impedance.

Because of this, per-unit calculations in power systems are almost always carried out on a 3-phase kVA / line-to-line kV basis, regardless of connection.

Single-Phase vs Three-Phase Base Values — Side by Side

Quantity	Single-Phase	Three-Phase
Base voltage, kV_B	Phase (line-to-neutral) voltage	Line-to-line voltage
Base power, kVA_B	Per-phase kVA	Total 3-phase kVA
Base current, I_B	kVA_B / kV_B	kVA_B / ($\sqrt{3} \times kV_B$)
Base impedance, Z_B	$(kV_B)^2 \times 1000 / kVA_B$	$(kV_B)^2 \times 1000 / kVA_B$ — same form
Per-unit impedance	Actual Z / Z_B	Actual Z / Z_B — same form

Key takeaway: the base-impedance and per-unit-impedance formulas are identical in structure for 1-ϕ and 3-ϕ systems — only the meaning of kV_B and kVA_B changes.

Change of Base

Nameplate p.u. impedance must be converted to the system's common base before use

Equipment is rated in % / p.u. impedance on its OWN base (kVA, kV). A network study picks ONE common base for everything. Convert using:

$$Z_{pu,new} = Z_{pu,old} \times \frac{kVA_{new}}{kVA_{old}} \times \frac{(kV_{old})^2}{(kV_{new})^2}$$

Doubling the base kVA doubles the p.u. impedance on the new base.

$$\left(\frac{Z_{pu,\text{new}}}{Z_{pu,\text{old}}}\right)\left(\frac{V_{\text{new}}}{V_{\text{old}}}\right)^2 = \frac{S_{\text{new}}}{S_{\text{old}}}$$

A transformer's p.u. ipedance is identical whether referred to its primary or secondary side.

$$P = VI$$

$$I = \frac{V}{Z}$$

$$P = V\left(\frac{V}{Z}\right)$$

$$P = \frac{V^2}{Z}$$

$$\boxed{PZ = V^2}$$

$$P_{\text{new}} Z_{\text{new}} = V_{\text{new}}^2$$

$$P_{\text{old}} Z_{\text{old}} = V_{\text{old}}^2$$

Now divide **new by old**:

$$\boxed{\frac{P_{\text{new}}}{P_{\text{old}}} \frac{Z_{\text{new}}}{Z_{\text{old}}} = \left(\frac{V_{\text{new}}}{V_{\text{old}}}\right)^2}$$

$$Z_{pu} = \frac{Z_{\text{actual}}}{Z_{\text{base}}}$$

so the PU ratio is the **inverse of the impedance-base ratio**:

$$\frac{Z_{pu,\text{new}}}{Z_{pu,\text{old}}} = \frac{Z_{B,\text{old}}}{Z_{B,\text{new}}}$$

Hence,

$$\boxed{\frac{Z_{pu,\text{new}}}{Z_{pu,\text{old}}} = \frac{P_{\text{new}}}{P_{\text{old}}} \left(\frac{V_{\text{old}}}{V_{\text{new}}}\right)^2}$$

$$\left(\frac{Z_{pu,\text{new}}}{Z_{pu,\text{old}}}\right) \left(\frac{V_{\text{new}}}{V_{\text{old}}}\right)^2 = \frac{S_{\text{new}}}{S_{\text{old}}}$$

Worked Example

Single-phase transformer — leakage reactance to per-unit

Given

- Single-phase transformer: 20 kVA, 2000/200 V
- Leakage reactance referred to the low-voltage (200 V) side: X = 0.8 Ω
- Find: per-unit leakage reactance

Base values (LV side)

$kV_B = 0.2$ kV, $kVA_B = 20$ kVA

$Z_B = (0.2)^2 \times 1000 / 20 = 2\ \Omega$

Solution

$$Z_{pu} = \frac{\text{Actual } x}{Z_B}$$

$$= 0.8 / 2$$

$$\mathbf{X_{pu} = 0.4\ p.u.}$$

Check on the HV side: $kV_B = 2$ kV, so $Z_B = (2)^2 \times 1000/20 = 200\ \Omega$. Referred to HV, $X = 0.8 \times (2000/200)^2 = 80\ \Omega$, giving p.u. reactance = 80/200 = 0.4 p.u. — identical, confirming p.u. impedance is independent of side.

Base Impedance

$$Z_B = \frac{(kV_B)^2 \times 1000}{kVA_B} \quad \textit{ohms}$$

Base values (LV side)

$kV_B = 0.2$ kV, $kVA_B = 20$ kVA

$Z_B = (0.2)^2 \times 1000 / 20 = 2\ \Omega$

$$Z_{pu} = \frac{\text{Actual } x}{Z_B}$$

$$= 0.8 / 2$$

$$\mathbf{X_{pu} = 0.4\ p.u.}$$

Principle of Referring Impedance Across a Transformer

• Referring an impedance means replacing it with an equivalent impedance on the other side of the transformer.

• The referred impedance produces the same power effect **(Actual and reactive)** when carrying the corresponding current on that side.

For resistance:

$$\text{Original copper loss} = I_1^2 R_1$$

When R_1 is referred to the secondary:

$$I_2^2 R_1' = I_1^2 R_1$$

$$\text{Since } I_2 = aI_1, \text{ where } a = N_1/N_2,$$

$$R_1' = R_1/a^2$$

Same power effect → Different V and I → Impedance changes by a^2

Per unit impedance referred to the HV Side

$$\text{on the HV side: } kV_B = 2\ kV,$$
$$\text{so } Z_B = (2)^2 \times 1000/20 = 200\ \Omega$$

Since we move from the LV side to the HV side, the value of k = 2000/200

X referred to HV = X referred to LV X k^2

$$X = 0.8 \times (2000/200)^2 = 80\ \Omega$$
$$\text{p.u. reactance} = 80/200 = 0.4\ \text{p.u}$$

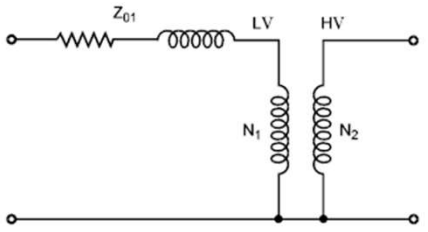
Eliminating the Transformer — Referring the Impedance

The leakage impedance can be referred to either side — whichever matches your chosen base voltage

1. The impedance can be referred either way across the winding

$$Z_{02} = Z_{01} \times (N_2/N_1)^2$$ *primary → secondary*

$$Z_{01} = Z_{02} \times (N_1/N_2)^2$$ *secondary → primary*



2.10. *Simplified Equivalent Circuit of a Single-Phase Two Winding Transformer*

Both equations describe the same physical impedance. Use whichever direction refers the impedance to the side where you have already fixed a base voltage.

2. Convert the referred impedance to per-unit, on that same side's base

$$Z_{pu} = \frac{\text{Referred impedance } (\Omega)}{\text{Base impedance on that same side}}$$

Eliminating the Transformer — Same Result, Either Direction

Once bases follow the transformer ratio, the referral direction no longer matters

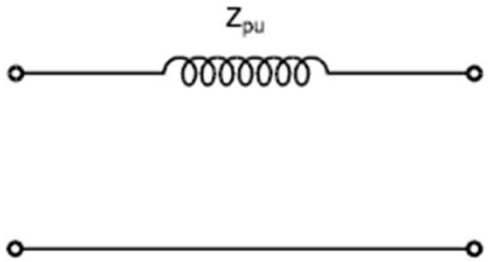
3. Choose base kV in the transformer's own turns ratio

$$kV_{B2} / kV_{B1} = N_2 / N_1$$

Crossing the transformer, scale the base voltage exactly as the transformer itself scales actual voltage — this rule applies the same way whether you referred Z to the primary or the secondary.

4. Either direction gives the same per-unit value

$$Z_{01,pu} = Z_{02,pu} = Z_{pu}$$



Whether you refer the impedance to the primary side or the secondary side, converting to per-unit on the matching base gives the identical number. The turns ratio N_1:N_2 cancels out completely — a single per-unit impedance Z^{pu} represents the transformer, so it can be dropped from the diagram.

1 Simplified calculations

No repeated √3 or transformer-ratio bookkeeping once everything is on one base.

2 Meaningful comparison

Machine parameters (X'', X', X) fall in a narrow, predictable p.u. range regardless of physical rating — erroneous data stand out.

3 Transformers vanish from the diagram

With the correct base change across each winding, ideal transformers disappear from the per-unit impedance diagram entirely.

4 One base serves the whole network

Once chosen, the same base kVA is used system-wide; only base kV changes (by transformation ratio) across each transformer.

Example 2.6. Determine the per unit impedance of a transmission line having an impedance of $(30 + j\,110)\ \Omega$ on 100 MVA and 132 kV base voltage.

Solution : Impedance of line, $Z = (30 + j\,110)\ \Omega$

Base kVA, $kVA_B = 100$ MVA or 100,000 kVA

Base kV, $kV_B = 132$ kV

$$\text{Per unit impedance, } Z_{pu} = \frac{Z \times kVA_B}{(kV_B)^2 \times 1{,}000}$$

$$= \frac{(30 + j110) \times 100{,}000}{(132)^2 \times 1{,}000}$$

$$= (0.172 + j\,0.631)\ \text{pu}$$ **Ans.**

Example 2.7. A 30 MVA, 11 kV generator has a reactance of 0.2 pu referred to its ratings as bases. Determine the per unit reactance when referred to base kVA of 50,000 kVA and base kV of 33 kV.

Solution : Old base kVA = 30,000 kVA

Old base kV = 11 kV

Old per unit impedance,

$$Z_{pu\ old} = 0.2 \text{ pu}$$

New base kVA = 50,000 kVA

New base kV = 33 kV

New per unit impedance,

$$Z_{pu\ new} = Z_{pu\ old} \times \frac{\text{New base kVA}}{\text{Old base kVA}} \times \left(\frac{\text{Old base kV}}{\text{New base kV}} \right)^2$$

$$= 0.2 \times \frac{50{,}000}{30{,}000} \times \left(\frac{11}{33} \right)^2 = 0.037 \text{ pu } \textbf{Ans.}$$

A balanced **3-phase system** has:

$$S = 30\ \text{MVA}, \qquad V_L = 12\ \text{kV}, \qquad Z = 1.5 + j2.0\ \Omega/\text{phase}$$

Choose the initial bases as:

$$S_{B1} = 50\ \text{MVA}, \qquad V_{B1} = 10\ \text{kV}$$

Find:

1. Base current and base impedance
2. Per-unit voltage, current and impedance
3. The per-unit impedance on a new base of **100 MVA, 20 kV**

1. Initial base quantities

For a 3-phase system,

$$I_{B1} = \frac{S_{B1}}{\sqrt{3}V_{B1}} = \frac{50}{\sqrt{3}(10)}$$

$$\boxed{I_{B1} = 2.887\ \text{kA}}$$

Base impedance:

$$Z_{B1} = \frac{V_{B1}^2}{S_{B1}} = \frac{10^2}{50}$$

$$\boxed{Z_{B1} = 2\ \Omega}$$

Voltage

$$V_{pu} = \frac{V}{V_{B1}} = \frac{12}{10}$$

$$\boxed{V_{pu} = 1.2}$$

Current

Actual line current is

$$I = \frac{S}{\sqrt{3}V} = \frac{30}{\sqrt{3}(12)} = 1.443 \text{ kA}$$

Therefore,

$$I_{pu} = \frac{1.443}{2.887}$$

$$\boxed{I_{pu} = 0.5}$$

Impedance

$$Z_{pu,1} = \frac{Z}{Z_{B1}} = \frac{1.5 + j2.0}{2}$$

$$\boxed{Z_{pu,1} = 0.75 + j1.0 \, pu}$$

Notice also that

$$|Z| = \sqrt{1.5^2 + 2^2} = 2.5\Omega$$

so

$$|Z|_{pu} = \frac{2.5}{2} = 1.25 \, pu$$

3. Change the base

Now choose:

$$S_{B2} = 100\ \text{MVA}, \qquad V_{B2} = 20\ \text{kV}$$

Use

$$Z_{pu,2} = Z_{pu,1} \left(\frac{S_{B2}}{S_{B1}}\right) \left(\frac{V_{B1}}{V_{B2}}\right)^2$$

Substitute:

$$Z_{pu,2} = (0.75 + j1.0) \left(\frac{100}{50}\right) \left(\frac{10}{20}\right)^2$$

$$= (0.75 + j1.0)(2)(0.25)$$

Therefore,

$$\boxed{Z_{pu,2} = 0.375 + j0.50\ pu}$$